# CAST Science Investigation: Projectile Motion Data Analysis

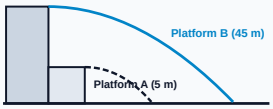


**UNIT 2 WRAP-UP • DATA TABLES, TRENDS & SCIENTIFIC ARGUMENTS (FORM B)**

**Name:** _____ **Per:** _____ **Date:** _____

## EXPERIMENTAL SETUP: TWO-PART PROJECTILE INVESTIGATION

Gravity: g = 10 m/s²




- **Investigation 1 (Two Launch Heights):** Balls are fired horizontally ($\theta$ = **0°**) from **Platform A** ($h$ = **5 m**, $t$ = **1.0 s**) and **Platform B** ($h$ = **45 m**, $t$ = **3.0 s**).
- **Investigation 2 (Angled Ground Launches):** On level ground at constant $v_o$ = **20 m/s**, students test five launch angles (**15°–75°**).

✏️ **Scan Card Instructions (Right):** Write & bubble your **6-digit Student ID** clearly inside the box on the right. **Leave the SCORE bubbles blank** for your teacher to bubble & scan!

**CAST Projectile Data Analysis** **MAX SCORE: 10**

Unit 2 Wrap-Up Investigation (Form B)

Bubble your 6-digit Student ID. Leave SCORE blank for teacher.

**STUDENT ID**

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| | | | | | |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 | 9 | 9 |

**SCORE**

☐☐ / 10

| | |
|---|---|
| 0 | 0 |
| 1 | 1 |
| 2 | 2 |
| 3 | 3 |
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## INVESTIGATION 1: HORIZONTAL LAUNCH SPEED ($V_{OX}$) VS. LANDING RANGE (ΔX)

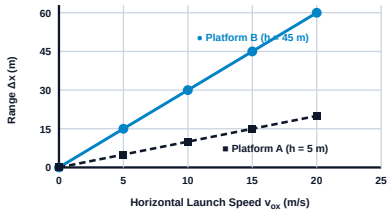
Platform A (h = 5 m) vs. Platform B (h = 45 m)

**Table 1: Horizontal Launch Range from Two Heights**

| Horizontal Speed $v_{ox}$ (m/s) | Platform A Range ($h$ = 5 m) $\Delta x_A$ (m) [■ Dashed] | Platform B Range ($h$ = 45 m) $\Delta x_B$ (m) [● Solid] |
|---|---|---|
| 0 | 0 | 0 |
| 5 | 5 | 15 |
| 10 | 10 | 30 |
| 15 | 15 | 45 |
| 20 | 20 | 60 |

📐 **Physics Connection:** On a graph of **Range ($\Delta x$)** vs. **Horizontal Speed ($v_{ox}$)**, the **slope ($\Delta y / \Delta x$)** equals the projectile's **flight time ($t$)** in seconds!

**Graph 1: Landing Range (Δx) vs. Horizontal Speed ($v_{ox}$)**



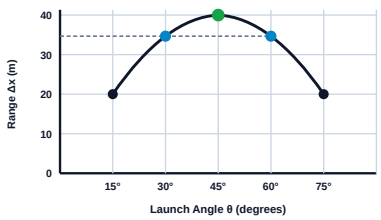

## INVESTIGATION 2: LAUNCH ANGLE (Θ) VS. VELOCITY COMPONENTS, HANG TIME & RANGE

Launch Speed: $v_o$ = 20 m/s

**Table 2: Angled Ground Launch Data ($v_o$ = 20 m/s)**

| Angle $\theta$ (°) | Horiz. Speed $v_{ox}$ (m/s) | Initial Vert. Speed $v_{oy}$ (m/s) | Hang Time $t_{hang}$ (s) | Landing Range $\Delta x$ (m) |
|---|---|---|---|---|
| 15° | 19.3 | 5.2 | 1.04 | 20.0 |
| 30° | 17.3 | 10.0 | 2.00 | 34.6 |
| 45° | 14.1 | 14.1 | 2.83 | 40.0 |
| 60° | 10.0 | 17.3 | 3.46 | 34.6 |
| 75° | 5.2 | 19.3 | 3.86 | 20.0 |

**Graph 2: Landing Range (Δx) vs. Launch Angle (θ)**



 **Formulas:** $\Delta x = v_{ox} \times t$ | $t_{hang} = 2v_{oy} / g$ | $g = 10\ m/s^2$

## PART 1: READING MULTI-SERIES GRAPHS & CLASSIFYING TRENDS (2 POINTS)

Investigation 1 & 2

☐ **(1) READ GRAPH** Use **Table 1 and Graph 1** to compare the two launch platforms when Horizontal Speed is $v_{ox}$ = **15 m/s**:

• Platform A Range ($\Delta x_A$) = _____ **m** • Platform B Range ($\Delta x_B$) = _____ **m** • How many times farther does Platform B launch the ball? _____ **× farther**

☐ **(2) CLASSIFY** Compare the mathematical shape of **Graph 1** vs. **Graph 2** (circle one in each bracket):

• **Graph 1 ($\Delta x$ vs. $v_{ox}$):** Shows a **[ linear direct relationship / non-linear symmetric arch ]** with a constant positive slope.

• **Graph 2 ($\Delta x$ vs. $\theta$):** Shows a **[ linear direct relationship / non-linear symmetric arch ]** that peaks at _____ °.

## PART 2: CALCULATING GRAPH SLOPE & EXTRAPOLATING (4 POINTS)

Stacked Math Required

### ☐ (3A) CALCULATE SLOPE OF PLATFORM B LINE (•)

**Use $v_{ox}$ = 0 m/s to $v_{ox}$ = 20 m/s on Graph 1:**

**Slope = $\Delta x_B \div v_{ox}$**

**Slope = _____ m ÷ _____ m/s = _____ s**

**Physical Meaning: What does this 3 s slope represent?**

_____

### ☐ (3B) PREDICT RANGE AT $V_{OX}$ = 25 M/S

**Calculate how far the ball lands from Platform B ($t$ = 3 s) if launched horizontally at $v_{ox}$ = 25 m/s:**

**$\Delta x_B = v_{ox} \times t$**

**$\Delta x_B$ = _____ m/s × _____ s = _____ m**

**From Platform A ($t$ = 1 s) at 25 m/s: $\Delta x_A$ = _____ m**

## PART 3: CAST CONSTRUCTED RESPONSE — CLAIM, EVIDENCE & REASONING (4 POINTS)

Rubric: Claim (1) + Evidence (2) + Reasoning (1)

🧪 **CAST Investigation Prompt:** During Investigation 2, a student argues: *"A launch angle of 30° and a launch angle of 60° cannot possibly land at the same distance because the 60° shot stays in the air much longer."*
**Evaluate the student's argument using quantitative evidence from Table 2 and the formula $\Delta x = v_{ox} \times t_{hang}$.**

💬 **Optional CAST Sentence Starters:** • ***Claim:*** *"The student's argument is incorrect because 30° and 60° are complementary angles that both land at..."*
• ***Evidence:*** *"According to Table 2, at 30° the horizontal speed is ___ m/s and hang time is ___ s (Range = ___ m), whereas at 60°..."*
• ***Reasoning:*** *"Because range equals $v_{ox} \times t_{hang}$, the 30° launch trades faster ___ for shorter ___, while 60°..."*

☐ **ZONE A — CLAIM (1 pt): State whether 30° and 60° land at the same distance and identify that distance.**

☐ **ZONE B — EVIDENCE (2 pts): Cite specific numbers & units from Table 2 for BOTH 30° and 60° ($v_{ox}$, $t_{hang}$, and $\Delta x$).**

☐ **ZONE C — REASONING (1 pt): Explain WHY multiplying $v_{ox} \times t_{hang}$ gives the exact same range for both angles, and state why 45° is the maximum.**